Questions by topic: Ecological monitoring & new technologies - Set 1 (65 marks)

Answer all questions in the spaces provided.

0 1 A conservation team is planning to protect a population of an endangered antelope in a reserve.

0 1 . 1 Explain why each of the following features of the population should be measured before a conservation strategy is decided.

[8 marks]

Population size .....

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Distribution .....

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Survival rate .....

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Age structure .....

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0 1 . 2 Describe how the population size of the antelope could be estimated.

[2 marks]

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0 2 . 1 Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

Satellite / radio / GPS tracking .....

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DNA databases and eDNA .....

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Image recognition software .....

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Acoustic monitoring / sonograms .....

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0 2 . 2 Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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0 3 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 3 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 4 . 1 Which technique would be most suitable for detecting whether great crested newts are present in a pond without catching them? Tick **one** box.

[1 mark]

A Environmental DNA (eDNA) analysis ☐

B Radio tracking ☐

C Lincoln index (mark-release-recapture) ☐

D Leg rings ☐

0 4 . 2 Explain why geolocator tags, rather than radio or GPS transmitters, are used to track the migrations of small birds.

[2 marks]

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0 4 . 3 Explain how DNA databases could help to control the illegal trade in elephant ivory.

[2 marks]

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0 5 . 1 Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]

Table 1

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

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0 5 . 2 Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

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0 5 . 3 Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.

[2 marks]

0 5 . 4 Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

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0 6 . 1 Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

0 6 . 2 Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

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0 6 . 3 Explain how auditory monitoring can be used to study bats.

[2 marks]

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0 6 . 4 Explain **three** assumptions of the mark-release-recapture method that may not be met.

[3 marks]

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0 7 Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

0 7 . 1 Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer ☐ %

0 7 . 2 Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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0 7 . 3 Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

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0 7 . 4 Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

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Extra space

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END OF QUESTIONS